Figures — Model-free multi-agent reinforcement learning simulation of the influence of empathy on the sustainability of agents' consumption in a common-pool resource dilemma

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Study 1: Factorial Manipulation of Observation and Empathic Reward

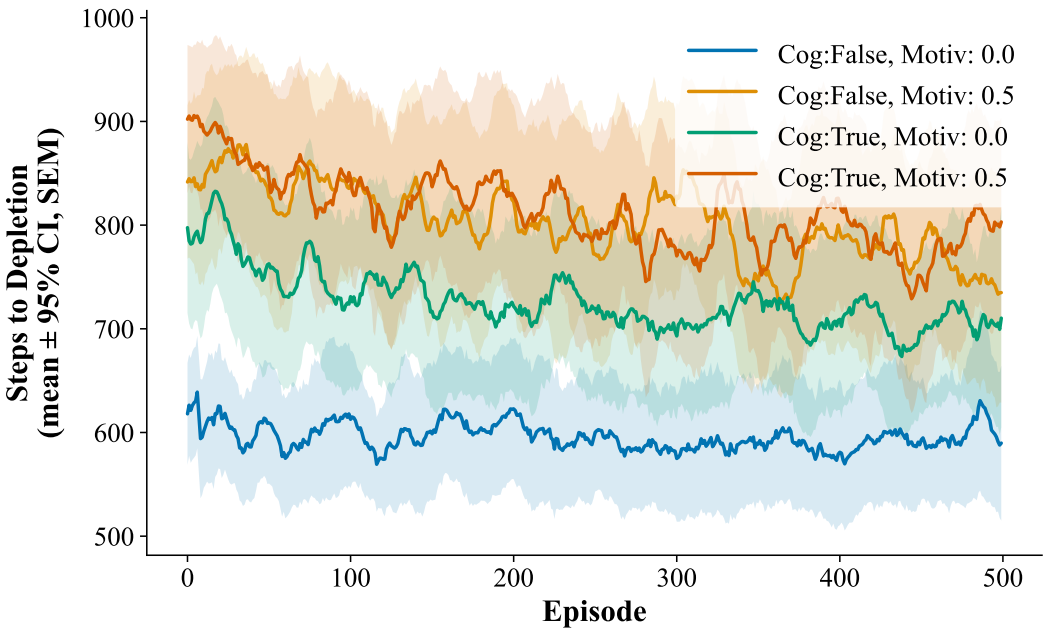


Figure 1: Episode length as a function of empathic properties across episodes. *Cog*: ability to perceive others' emotions; *Motiv*: empathic motivation.

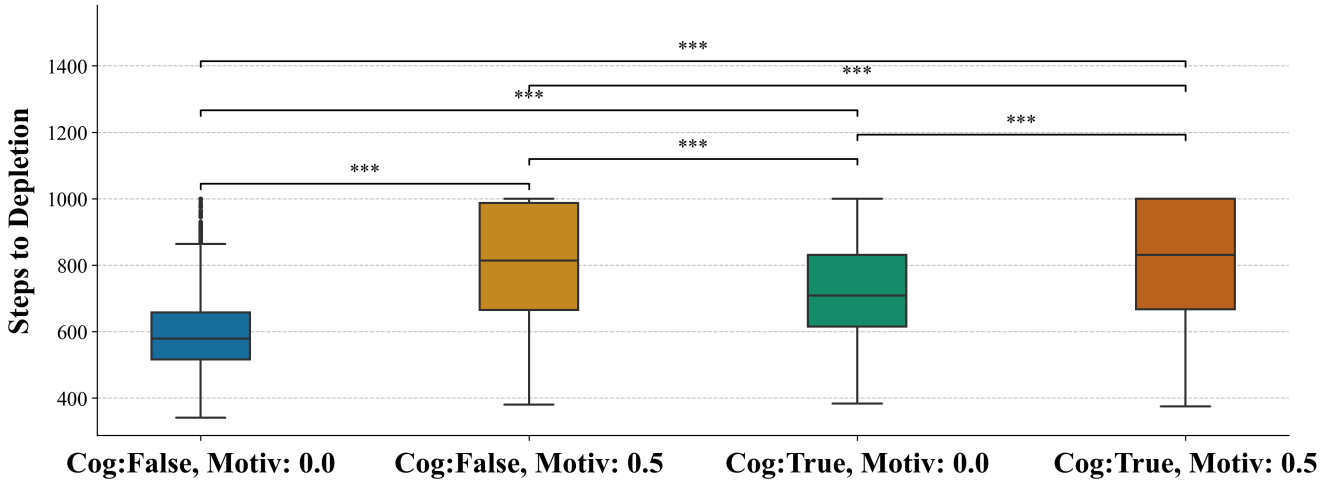


Figure 2: Episode length as a function of empathic properties, aggregated across episodes.

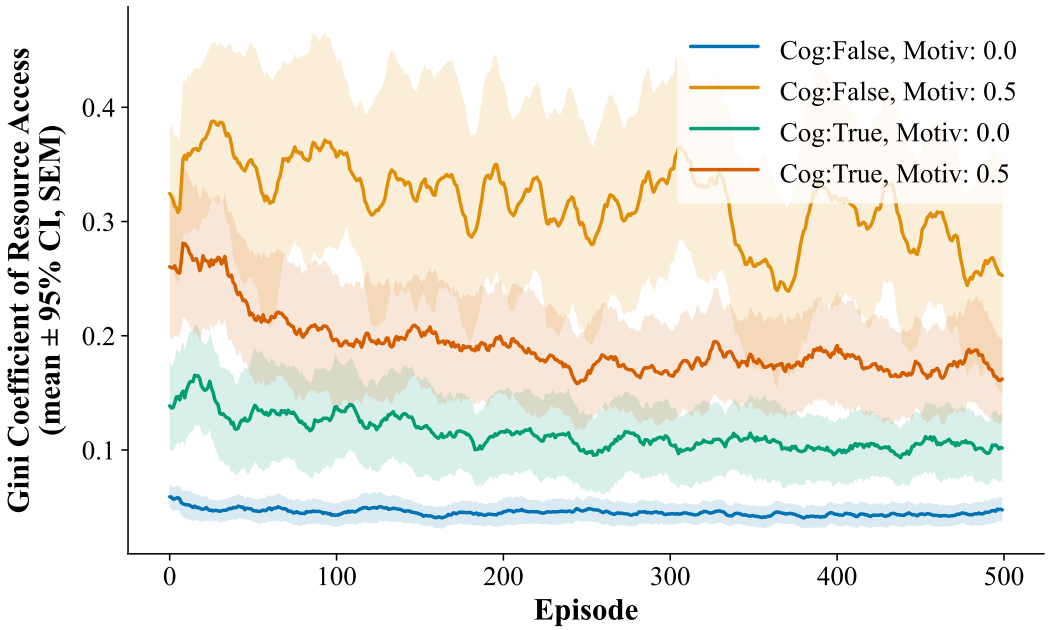


Figure 3: Fairness (Gini coefficient of resource access) as a function of empathic properties across episodes.

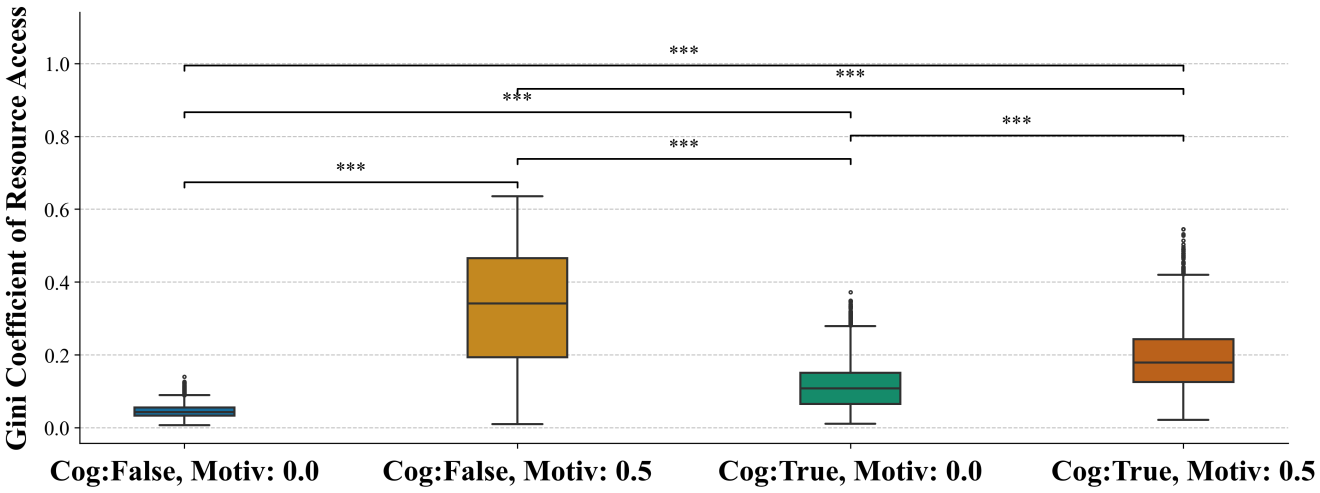


Figure 4: Fairness (Gini coefficient of resource access) as a function of empathic properties, aggregated across episodes.

Study 2: Graded Empathic Weight

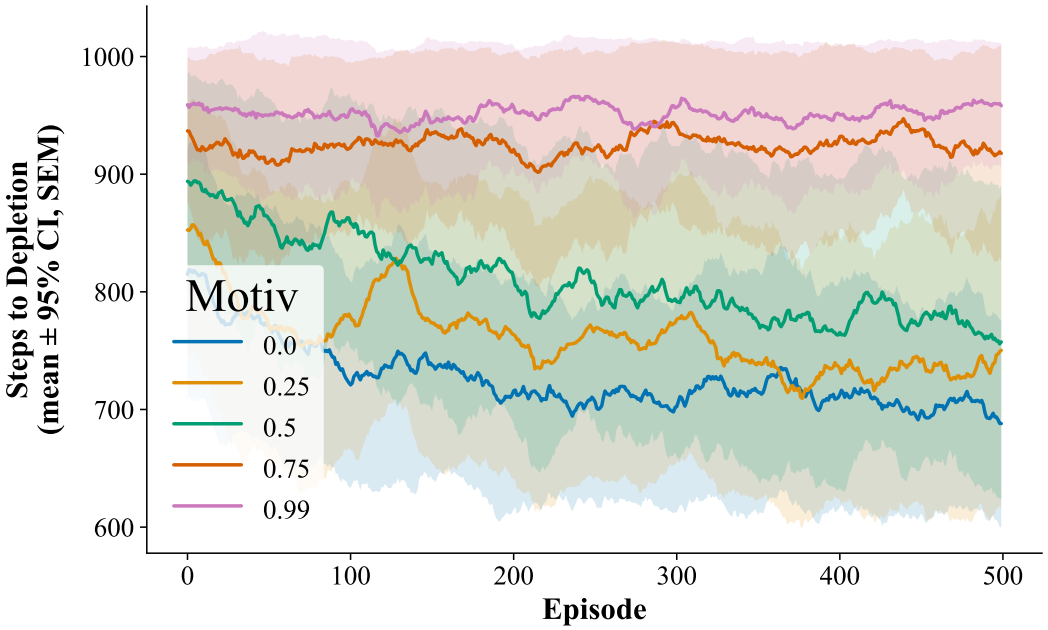


Figure 5: Episode length as a function of the degree of altruism (α_e) across episodes.

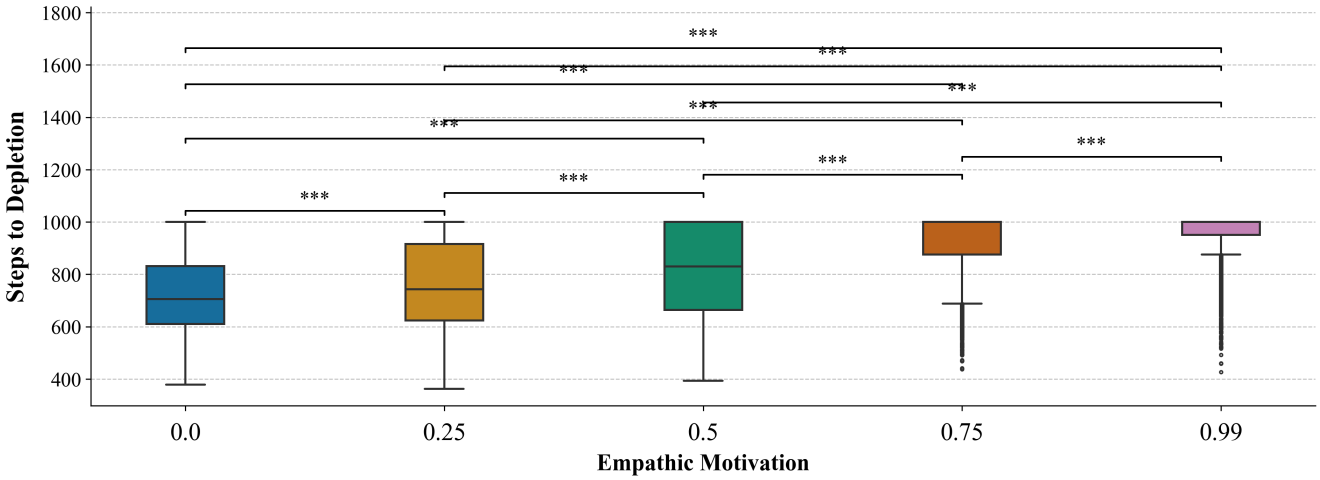


Figure 6: Episode length as a function of the degree of altruism (α_e), aggregated across episodes.

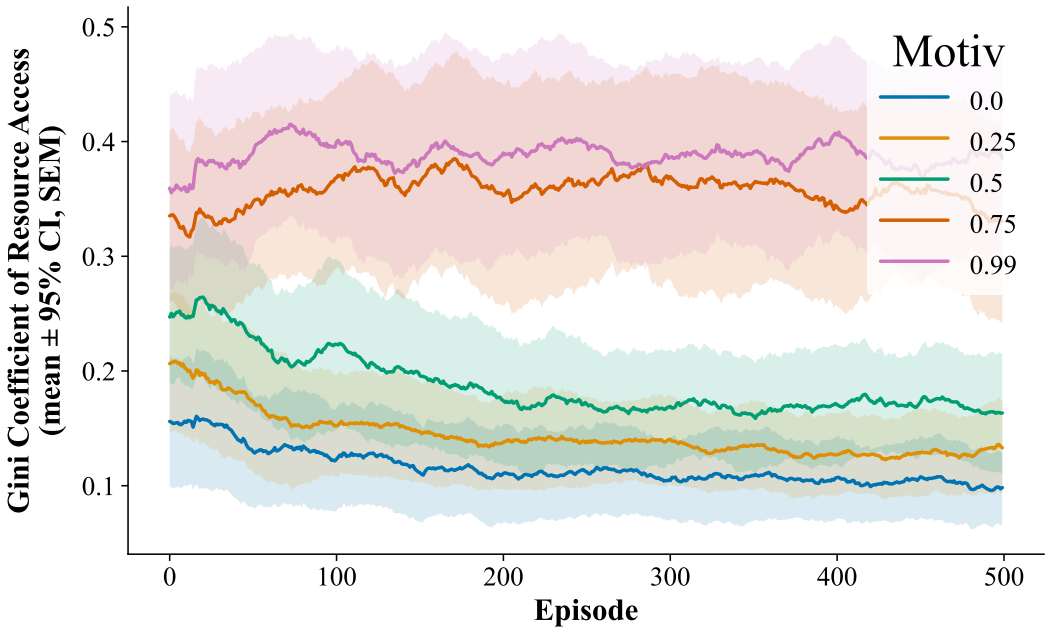


Figure 7: Fairness (Gini coefficient of resource access) as a function of the degree of altruism (α_e) across episodes.

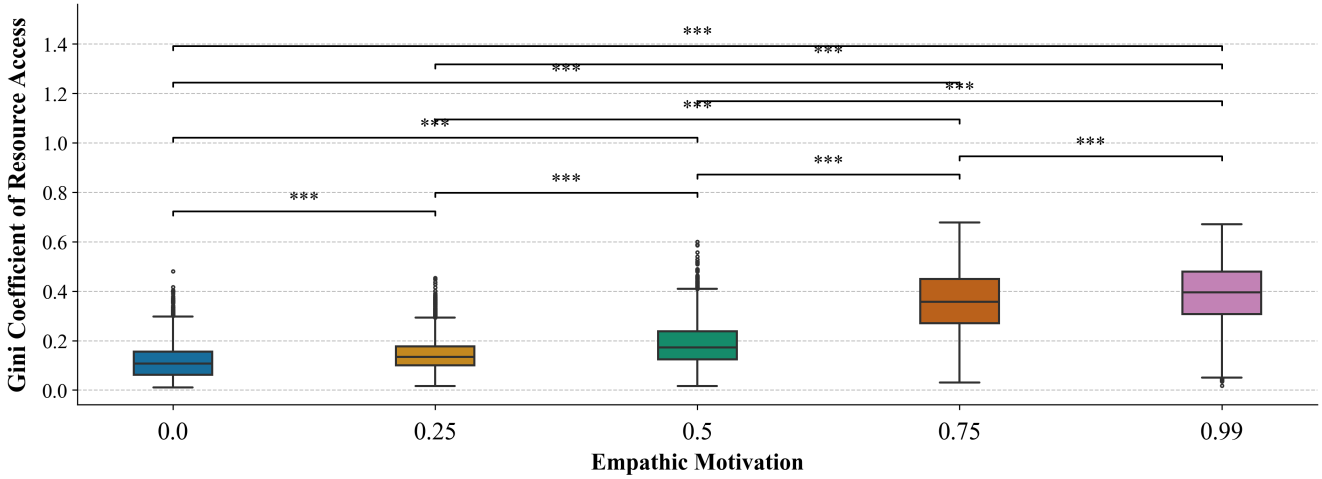


Figure 8: Fairness (Gini coefficient of resource access) as a function of the degree of altruism (α_e), aggregated across episodes.